

# Cross-Domain Synthetic Pre-Training in Quantum Continual Learning: An Empirical Characterization

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## Abstract

Sequential training of variational quantum classifiers produces interference between tasks, a failure mode known as catastrophic forgetting. Memory-based remedies conflict with the resource constraints of near-term quantum devices, which motivates initialization strategies that leave no memory footprint after training. This study characterizes one such strategy, where a hybrid classical-quantum model is pre-trained on a synthetic Gaussian source that shares no semantic content with the target sequence, and the resulting parameters initialize the sequential phase. A controlled factorial re-run of 360 training runs separates the contribution of the initialization from that of the optimization regime, and a benchmark campaign of 290 further runs extends the evaluation to class-incremental sequences of up to five tasks, qubit counts from four to eight, circuit depths from two to four and data budgets from five hundred to twelve thousand samples, under an ideal simulation and a calibrated device noise model, with paired tests, effect sizes and bootstrap intervals throughout. The controlled evidence relocates the effect that the original protocol ascribed to the initialization. The apparent reduction of the forgetting drop is carried by the learning-rate regime (about five points,  $p < 0.001$ ), the synthetic initialization produces no significant change under matched schedules, and the null reading is stable across benchmarks, scales and budgets. Mechanism probes show that the prior places the circuit in a far more active gradient region (roughly three orders of magnitude in the first layer) without a retention gain, so gradient signal and continual-learning performance dissociate and the separation between arms lies in the starting point rather than in a loss barrier. Rehearsal is the only evaluated remedy that improves average accuracy on the standard class-incremental benchmarks (+15.5 points,  $p < 0.001$ ), at a quantified cost in plasticity, and synaptic intelligence is the only weight-regularization baseline that improves over the plain model (+4.1 points,  $p = 0.007$ ) while cutting forgetting by an

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order of magnitude. The results are framed as a controlled characterization of cross-domain pre-training for quantum continual learning, not as a new method, and the limitations of the current scale are stated explicitly.

*Keywords:* quantum machine learning, continual learning, catastrophic forgetting, variational quantum circuits, transfer learning

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## 1. Introduction

Variational quantum circuits have become the workhorse of quantum machine learning in the noisy intermediate-scale quantum regime, where the reliable circuit size is limited by gate and readout errors [1, 2]. A variational circuit encodes classical data into a quantum state and updates rotation angles through a classical optimizer [3, 4], which makes the resulting model a candidate for classification tasks in settings where the data arrives as a stream rather than as a single static dataset [5]. When such a circuit is trained on a sequence of tasks without access to previous data, the updates for a new task overwrite the parameters that encoded earlier tasks, and the accuracy on the earlier tasks collapses. This phenomenon, catastrophic forgetting, has been characterized in classical networks [6, 7] and has recently been observed on superconducting hardware for quantum classifiers [8].

These limits have motivated two families of remedies in the classical literature. Regularization-based methods penalise changes to parameters that were important for previous tasks, with elastic weight consolidation as the canonical example [9] and with alternatives that estimate importance online [10]. Replay-based methods interleave stored samples or stored predictions with the current task, from plain experience replay [11] to the logit-distillation variants that preserve the soft decision boundaries of previous tasks [12]. Both families carry costs that sit badly with the constraints of near-term quantum devices. Regularization requires a Fisher information estimate whose storage and computation grow with the number of parameters, and replay requires a memory buffer whose size grows with the sequence [13, 14]. Recent work has shown that replay can be transferred to quantum-classical heads without destabilising training [15], which raises the question of what can be done when even a small buffer is not affordable.

In parallel, quantum transfer learning has developed the machinery to reuse knowledge across domains: a frozen classical network extracts compact features and a variational circuit is fine-tuned on them, which allows small quantum circuits to operate on high-dimensional inputs [16–18]. The transfer-learning literature in the classical setting has long recognised that pre-training on data unrelated to the target domain still produces features that transfer well [19, 20]. Whether the same logic applies to a variational circuit in a sequential setting, and what property of the resulting initialization protects later tasks, remains open. The observation that layerwise initialization strategies stabilise the training of quantum circuits [21] indicates that the starting point of a variational circuit is not a neutral design choice.

These constraints frame the question of this study: does pre-training a hybrid quantum-classical classifier on a synthetic source that shares no semantic content with the target tasks produce an initialization that reduces forgetting during later sequential training, and under which conditions does the effect survive? The study answers the question for a four-qubit hybrid model across circuit topologies, noise profiles, data budgets and qubit counts, with a statistical protocol built on paired comparisons over seeds. The contributions are the following. First, the cross-domain synthetic pre-training setup is formalised for a sequential quantum classifier, with the source, the task sequence and the evaluation protocol stated precisely. Second, a controlled factorial design separates the effect of the initialization from the effect of the optimization regime and shows that the apparent reduction of the forgetting drop, of about six points under the original protocol, is carried by the learning rate, while the initialization itself produces no significant change once rates are matched. Third, the null reading of the initialization is shown to be stable across class-incremental benchmarks of up to five tasks, data budgets from five hundred to twelve thousand samples and scaling grids over qubit counts and depths, and the comparative evaluation is reported throughout with absolute accuracies, paired tests, effect sizes and confidence intervals, which corrects two reporting defects of the earlier conference version of this study, namely the omission of absolute accuracies and a mislabeled retention table. Fourth, mechanism evidence based on gradient statistics, curvature and loss interpolation locates the role of the prior in the geometry of the starting point rather than in retention, which bounds what initialization alone can deliver and redirects attention to the components that do act, such as rehearsal and synaptic intelligence, whose behavior under the same protocol is quantified. The rest of the manuscript is organized as follows. Section 2 reviews the related literature. Section 3 fixes the scenario definitions and the metrics. Section 4 describes the pre-training methodology, the circuit architectures and the sequential protocol. Section 5 details the experimental setup. Section 6 reports the results, and Section 7 discusses their scope and limitations.

## 2. Related Work

### 2.1. Continual learning in classical networks

The sequential-learning problem was formalised by McCloskey and Cohen [6] and has since produced a taxonomy of strategies that spans regularization, replay and architectural design [7, 22, 23]. Elastic weight consolidation adds a quadratic penalty weighted by the diagonal of the Fisher information matrix [9], and synaptic intelligence replaces the Fisher estimate with an online accumulation of parameter importance [10]. Replay-based approaches store a small subset of previous samples and constrain the updates with them [11], and their distillation variants retain the soft outputs of the model on previous tasks [12, 24]. The distinction between task-incremental, domain-incremental and class-incremental scenarios [25] is central for the present study, because strategies that work under a task oracle do not transfer automatically to settings

where the task identity is unknown at inference time. Parameter-space regularization has been examined for quantum circuits [15], while the initialization of the model, which is the variable manipulated here, has received less attention in that setting.

## 2.2. Quantum transfer learning and quantum continual learning

Hybrid classical-quantum transfer learning freezes a pre-trained classical network and trains a variational circuit on its features, [16]. Extensions of this template have characterized the effect of device noise [17] and the behavior of different classical backbones [18]. Trained variational circuits are known to suffer from trainability problems, from barren plateaus in deep circuits [13, 26, 27] to noise-induced flattening of the landscape [14], and architectures with local connectivity have been shown to avoid part of these effects [28]. The continual case is more recent. A regularization-based quantum continual learning protocol has been reported on a superconducting processor, where an elastic-weight-consolidation penalty allowed a quantum classifier to retain knowledge across three sequential tasks [8]. Earlier work described unitary update rules for quantum continual learning in simulation [29]. Across these studies, the pre-training domain is treated as given, and the question of how the initialization itself shapes sequential retention is not isolated.

## 2.3. Research gaps

Table 1 places the present study next to representative prior work across the dimensions that matter for this analysis. Classical benchmarks report forgetting under full-data regimes, quantum studies focus on single-task transfer or on regularization, and the hardware realisation of quantum continual learning covers a short sequence with a regularization penalty. No prior study isolates the effect of a synthetic cross-domain initialization on sequential quantum classifiers, and none reports the comparison with paired statistics over seeds and per-seed confidence intervals. These two elements define the space of this study.

Table 1: Positioning against representative prior work. The four rightmost columns mark, respectively, whether the study reports hardware execution, more than one random seed, inferential statistics, and a scenario analysis beyond task-incremental evaluation. The marker n/a indicates aspects that do not apply to the study.

| Study                    | Setting   | Mechanism       | Sequence | HW  | Seeds | Stats | Scenario |
|--------------------------|-----------|-----------------|----------|-----|-------|-------|----------|
| Kirkpatrick et al. [9]   | Classical | EWC             | Multi    | n/a |       | n/a   | TIL      |
| Buzzega et al. [12]      | Classical | DER++           | Multi    | n/a |       | n/a   | TIL/CIL  |
| Mari et al. [16]         | Hybrid    | QTL             | Single   | n/a | n/a   | n/a   | n/a      |
| Zhang et al. [8]         | Quantum   | EWC             | 3 tasks  | yes | n/a   | n/a   | TIL      |
| Martín-Pérez et al. [15] | Hybrid    | Replay          | Multi    | n/a | yes   | yes   | TIL      |
| This study               | Hybrid    | Synthetic prior | 2 tasks  | n/a | yes   | yes   | TIL      |

### 3. Background and Problem Formulation

#### 3.1. Variational quantum classifiers

A variational quantum circuit prepares a state  $|\psi(x, \theta)\rangle$  by applying an encoding unitary that depends on the input  $x$  followed by a parameterised unitary  $U(\theta)$ , and produces predictions from the expectation values of a set of observables. Training minimizes a classical loss with respect to  $\theta$  through the parameter-shift rule [3] or through differentiable simulation. The expressibility and the entangling capability of the parameterised unitary depend on its topology [30, 31], and the trainability of the circuit depends on the same design choices [13, 28].

#### 3.2. Continual learning scenarios and metrics

A model that faces a sequence of tasks  $\{\mathcal{T}_1, \dots, \mathcal{T}_T\}$  observes the training data of task  $t$  while training on task  $t$  and, in the strictest form of the problem, keeps no access to the data of previous tasks. Two scenarios are distinguished after van de Ven et al. [25]. In task-incremental learning the task identity is known at test time and a shared head scores the corresponding decision boundary. In class-incremental learning the task identity is unknown and a single output layer must separate classes drawn from every task seen so far. The present study focuses on the task-incremental scenario and the analysis of class-incremental behavior is left to the extended protocol described in Section 7.

The evaluation uses the standard continual-learning metrics [11, 22]. Let  $A_{t,j}$  be the test accuracy on task  $j$  measured after training through task  $t$ . The average accuracy and the average forgetting over the sequence are

$$\text{AA} = \frac{1}{T} \sum_{j=1}^T A_{T,j}, \quad \text{AF} = \frac{1}{T-1} \sum_{j=1}^{T-1} (\max_{t \leq T} A_{t,j} - A_{T,j}), \quad (1)$$

and the backward transfer is the signed change of each task accuracy between its completion and the end of the sequence,

$$\text{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} (A_{T,j} - A_{j,j}). \quad (2)$$

For the two-task protocol analyzed here, the quantity of interest is the forgetting drop on the first task

$$\Delta_A = \text{Acc}_A^{\text{init}} - \text{Acc}_A^{\text{final}}, \quad (3)$$

where  $\text{Acc}_A^{\text{init}}$  is the accuracy on Task A measured at the end of the Task A phase and  $\text{Acc}_A^{\text{final}}$  is the same accuracy measured after the Task B phase, on the frozen Task A test set. The relative retention  $r_A = \text{Acc}_A^{\text{final}} / \text{Acc}_A^{\text{init}}$  is reported together with  $\Delta_A$ , because the absolute drop alone rewards classifiers that never learned the first task.

Two properties fix the role of  $\Delta_A$  among these quantities. First, for a sequence of two tasks the peak accuracy of Task A is attained at the end of its

own phase, so the average forgetting of Eq. (1) and the backward transfer of Eq. (2) reduce to

$$\Delta_A = \text{AF} = -\text{BWT} \quad (T = 2), \quad (4)$$

and the three measures carry the same information at this sequence length.  $\Delta_A$  is used because it is the quantity reported by the earlier conference version of this study, and it embeds directly into the BWT and AF definitions as the sequence grows. Second,  $\Delta_A$  is bounded above by  $\text{Acc}_A^{\text{init}}$ , so a classifier that never acquires Task A cannot lose much and shows a small drop for the wrong reason. The drop is therefore read jointly with  $\text{Acc}_A^{\text{init}}$  and  $r_A$ , and Section 6 shows this failure mode in the topology comparison, where the least accurate architecture shows the smallest drop.

## 4. Methodology

### 4.1. Cross-domain synthetic pre-training

The methodology separates three domains. The source domain  $\mathcal{S}$  is a synthetic binary classification problem generated in four dimensions from two Gaussian clusters, with no semantic relation to the target tasks. The sequential domain contains Task A, a binary classification problem built from two classes of Fashion-MNIST [32], and Task B, a binary problem built from two classes of MNIST [33]. A hybrid model consists of a frozen feature extractor and a variational circuit head. For image tasks, a MobileNetV2 network pre-trained on ImageNet produces a feature vector that is reduced to four components by principal component analysis [34, 35]. For the synthetic source the four features are drawn directly from the Gaussian generator. All features are rescaled to the interval  $[0, \pi]$  to match the rotation domain of the encoding layer.

Pre-training minimises the cross-entropy loss of the hybrid model on the synthetic source for a fixed number of epochs and stores the resulting parameters  $\theta_0$ . The sequential phase initializes the circuit with  $\theta_0$  and trains on Task A and then on Task B, as summarised in Algorithm 1. Two initialization references complete the design: a scratch model, initialized at random, and a cross-domain model whose source features come from CIFAR-10 images passed through the same MobileNetV2 backbone [36], which separates the effect of the synthetic source from the effect of pre-training on real images.

*Working hypothesis..* The study tests the hypothesis that pre-training a parameterised circuit on generic synthetic data yields a more favourable initialization for subsequent sequential training than a random draw. Three arguments motivate the expectation. From the classical side, features learned on a source with no semantic relation to the target still transfer to it [19, 20], so the benefit of pre-training need not depend on domain overlap. From the quantum side, the initialization of a variational circuit is not neutral for its trainability: layerwise initialization stabilises the training of small circuits [21], and random draws in wide circuits concentrate in flat regions of the landscape [13]. From the design side, an isotropic Gaussian source with full-rank covariance activates every

rotation axis of the encoding layer, so the source optimization moves the parameters through a region where the circuit responds to structured inputs, instead of leaving them near the small-angle regime of a canonical random draw. The hypothesis yields three testable predictions: the prior reduces the forgetting drop without raising the accuracy ceiling of the target task. The gradient signal of the target loss at the pre-trained point is stronger than at random points, and the benefit does not require semantic overlap between source and target. The first two are tested in Sections 6 and 6.11, and the third is probed by the cross-domain image reference of Experiment 3.

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**Algorithm 1:** Cross-domain sequential protocol with synthetic pre-training.

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**Input** : Source data  $D_S$ , tasks  $\{D_A, D_B\}$ , learning rates  $\eta_S, \eta_A, \eta_B$ , epochs  $E_S, E_A, E_B$

**Output:**  $\theta_0$ , accuracies  $\{\text{Acc}_A^{\text{init}}, \text{Acc}_B, \text{Acc}_A^{\text{final}}\}$ , forgetting drop  $\Delta_A$

- 1 Sample  $\theta$  at random or load a pre-trained initialization ;
- 2 **if** *pre-training* **then**
- 3     Train the hybrid model on  $D_S$  with learning rate  $\eta_S$  for  $E_S$  epochs ;
- 4     Store  $\theta_0 \leftarrow \theta$  ;
- 5 Train the model with initialization  $\theta$  on  $D_A$  with  $\eta_A$  for  $E_A$  epochs ;
- 6 Evaluate  $\text{Acc}_A^{\text{init}}$  on the Task A test set ;
- 7 Train the same model on  $D_B$  with  $\eta_B$  for  $E_B$  epochs ;
- 8 Evaluate  $\text{Acc}_B$  on the Task B test set and  $\text{Acc}_A^{\text{final}}$  on the Task A test set ;
- 9 Return  $\Delta_A = \text{Acc}_A^{\text{init}} - \text{Acc}_A^{\text{final}}$  ;

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#### 4.2. Circuit architectures

Three topologies are evaluated on four qubits, all preceded by the same encoding block and followed by the same readout. The encoding applies a single-qubit rotation around the  $X$  axis to each wire,

$$|\psi_{\text{in}}(x)\rangle = \bigotimes_{i=0}^{n-1} R_X(x_i) |0\rangle, \quad (5)$$

where  $x \in [0, \pi]^n$  and  $n = 4$ . The readout measures the expectation of the Pauli-Z operator on the first two wires,

$$z_k = \langle \psi_{\text{out}} | Z_k | \psi_{\text{out}} \rangle, \quad k \in \{0, 1\}, \quad (6)$$

with  $|\psi_{\text{out}}\rangle = U(\theta) |\psi_{\text{in}}(x)\rangle$ . The two expectation values feed a fully connected classical layer that produces the binary logits  $\hat{y} = Wz + b$ , with  $W \in \mathbb{R}^{2 \times 2}$  and  $b \in \mathbb{R}^2$ , and the model is trained with categorical cross-entropy.

*Strongly entangling layers (SEL).* Each layer applies a general single-qubit rotation  $R(\alpha, \beta, \gamma)$  to every wire, followed by a shifted ring of CNOT gates

whose range depends on the layer index,

$$U_{\text{SEL}}(\theta) = \prod_{\ell=1}^L \left[ \prod_{q=0}^{n-1} R_q(\theta_{\ell,q,1}, \theta_{\ell,q,2}, \theta_{\ell,q,3}) \cdot \prod_{q=0}^{n-1} \text{CNOT}(q, (q + r_\ell) \bmod n) \right], \quad (7)$$

where  $r_\ell = (\ell - 1) \bmod (n - 1) + 1$  and the parameter tensor has shape  $(L, n, 3)$ , which gives 36 parameters for the configuration used here.

*Basic entangler (BE).* Each layer applies one rotation around the  $X$  axis per wire, followed by a ring of CNOT gates,

$$U_{\text{BE}}(\theta) = \prod_{\ell=1}^L \left[ \prod_{q=0}^{n-1} R_X(\theta_{\ell,q}) \cdot \prod_{q=0}^{n-1} \text{CNOT}(q, (q + 1) \bmod n) \right], \quad (8)$$

with a parameter tensor of shape  $(L, n)$ , which gives 12 parameters.

*Tree tensor network (TTN).* The tree pairs its wires hierarchically. At the first level, adjacent wires are paired and the information of each pair accumulates on the right wire, and the procedure repeats on the surviving wires until a single wire remains. Each block applies one rotation around  $Y$  and one rotation around  $Z$  followed by a CNOT,

$$U_{\text{TTN}}(\theta) = \prod_{\ell=1}^L \prod_{b=0}^{n-2} \text{CNOT}(i_b \rightarrow j_b) R_Z(\theta_{\ell,b,2})_{j_b} R_Y(\theta_{\ell,b,1})_{i_b}, \quad (9)$$

where  $(i_b, j_b)$  is the wire pair of block  $b$  in the hierarchical wiring. The parameter tensor has shape  $(L, n - 1, 2)$ , which gives 18 parameters for  $n = 4$  and  $L = 3$ . Two properties of this construction are relevant for the reading of the results. First, the accumulation of information proceeds towards the right end of the tree, so the subset of parameters that feeds the first two wires, where the readout of Eq. (6) is placed, is smaller than the nominal parameter count. Second, the blocks that act on subtrees disconnected from the readout wires contribute no gradient under this readout configuration, an effect measured in the mechanism analysis of Section 6.11. The comparison across topologies is therefore also a comparison across effective circuit sizes, and it is reported as such.

#### 4.3. Sequential training protocol

The sequential protocol trains a fresh model per configuration and seed. The pre-training phase runs for fifteen epochs on the source data. The Task A phase loads the pre-trained parameters  $\theta_0$  or a random initialization and trains on Task A for ten epochs, after which  $\text{Acc}_A^{\text{init}}$  is evaluated. The Task B phase continues the training on Task B for ten further epochs with the same optimizer state, and the protocol ends with the evaluation of  $\text{Acc}_B$  and  $\text{Acc}_A^{\text{final}}$ . No replay buffer and no parameter-space penalty are used in the reference protocol. The learning rate differs across phases and conditions by design of the original campaign, and the implications of that design choice are discussed with the results.

#### 4.4. Noise profiles

Every configuration is replicated under three noise profiles. The ideal profile propagates the circuit through the state-vector simulator. The two noisy profiles insert a single-qubit depolarising channel after every parameterised rotation, a two-qubit depolarising channel after every two-qubit gate, and a bit-flip channel on every measured wire to emulate readout error. The device profile borrows its parameters from the median calibration of the IBM Heron r2 generation, with a single-qubit depolarising rate of  $3 \times 10^{-4}$ , a two-qubit rate of  $3 \times 10^{-3}$  and a readout error of  $1.5 \times 10^{-2}$ . The legacy profile scales those rates towards earlier device generations, with a single-qubit rate of  $10^{-3}$ , a two-qubit rate of  $1.5 \times 10^{-2}$  and a readout error of  $3 \times 10^{-2}$ .

## 5. Experimental Setup

### 5.1. Datasets

Four data sources participate in the pipeline and Table 2 summarises their splits. The synthetic source generates two Gaussian clusters in four dimensions with the `make_classification` routine of scikit-learn. Task A uses classes 0 and 1 of Fashion-MNIST [32] and Task B uses classes 2 and 3 of MNIST [33]. The cross-domain reference extracts MobileNetV2 features from CIFAR-10 classes 2 and 3 [34, 36] and reduces them to four components. All image sources are reduced to four principal components fitted on the training split and rescaled to  $[0, \pi]$ .

Table 2: Sample counts per data source. The synthetic source is drawn directly in four dimensions. Image sources traverse a frozen MobileNetV2 backbone before the principal-component reduction.

| Source                          | Role             | Train | Test | Classes |
|---------------------------------|------------------|-------|------|---------|
| Synthetic (Gaussian clusters)   | Source           | 2000  | 500  | 2       |
| CIFAR-10 (MobileNetV2 features) | Source reference | 1000  | 200  | 2, 3    |
| Fashion-MNIST                   | Task A           | 1000  | 200  | 0, 1    |
| MNIST                           | Task B           | 1000  | 200  | 2, 3    |

### 5.2. Statistical protocol

Every configuration is executed with five independent random seeds and every reported quantity is the mean over seeds with its standard deviation. Comparisons between two configurations are paired by seed, which removes the seed-to-seed variance of the shared pipeline from the contrast. Each paired comparison reports the mean difference, a two-sided paired  $t$ -test, a Wilcoxon signed-rank test as a distribution-free check, the Cohen  $d_z$  effect size and a 95% bootstrap confidence interval of the mean difference estimated with twenty thousand resamples of the seed pairs. Families of comparisons are corrected with the Holm procedure. The analysis code and the per-seed payloads are

stored with the repository described in the data availability statement. The reported Wilcoxon probabilities illustrate a known limitation of five-seed campaigns, namely that the smallest attainable two-sided  $p$ -value of the signed-rank test with five pairs is 0.0625, which is why the paired  $t$ -test and the bootstrap interval carry the inferential weight here and why the extended campaign of Section 7 raises the seed count.

### 5.3. Implementation and resources

The pipeline is implemented in Python 3.11 with PennyLane 0.44.1 for the quantum simulation, PyTorch 2.10.0 for the hybrid model and the optimizer, scikit-learn 1.8.0 for the feature reduction, and NumPy 2.4.3. The ideal profile runs on the state-vector device with the adjoint differentiation method and the noisy profiles run on the density-matrix device with backpropagation. The optimizer is Adam with a batch size of 32 and a learning rate of 0.05 during pre-training. The five-seed campaign of the conference analysis run on the Hercules cluster of the Centro de Informática Científica de Andalucía, on standard nodes with four CPU cores and eight gigabytes of memory, as independent jobs of SLURM arrays. The controlled re-run of Experiment 1 reported in Section 6 was executed outside that cluster, because the cluster’s access VPN was unavailable at run time, on two local workstations: an Intel Core i5-8500 with six cores and an Intel Core i7-5820K with six cores and twelve hardware threads, with one single-threaded worker per run. The software stack of the controlled re-run is Python 3.12.3, PyTorch 2.14.0, PennyLane 0.45.1, scikit-learn 1.9.1 and NumPy 2.5.3, identical on both machines. Every stored run records the machine identity of the node that executed it (`hostname` together with the OS machine identifier), never a placeholder, so the per-node provenance of each measurement is auditable from the released payloads. The complete five-seed campaign over the three noise profiles consists of forty-five sequential trainings per experiment. The classical front-end of the image sources, namely the MobileNetV2 feature extraction and the principal-component reduction, is accounted separately as the reproduction cost of the data preparation and not as part of the quantum training loop. The end-to-end balance of the pipeline returns in Section 7, where the scope of the claims is delimited.

## 6. Results

### 6.1. Experiment 1: does the synthetic prior reduce the drop?

Table 3 reports the forgetting drop on Task A for the random initialization and for the synthetic prior across the three noise profiles, with the paired reduction and its bootstrap interval. Under the device profile, the random initialization loses  $72.90 \pm 2.01$  points of Task A accuracy after the Task B phase, and the synthetic prior reduces the loss to  $65.10 \pm 3.73$  points, a paired reduction of 7.80 points whose bootstrap interval  $[4.50, 12.30]$  excludes zero and whose paired test arrives at  $p = 0.028$ , which becomes  $p_{\text{Holm}} = 0.084$  when the three per-profile tests are corrected as one family. The ideal profile shows a

reduction of 7.00 points with a wider interval, and the legacy profile shows the largest nominal reduction, 9.30 points, with the widest spread, which reflects the larger seed variance of the noisiest condition. When the fifteen seed pairs of the three profiles are pooled, the reduction is 8.03 points with a bootstrap interval of  $[4.47, 11.63]$ , a paired test below 0.001, a signed-rank test at 0.003 and an effect size of  $d_z = 1.09$ . Read profile by profile, the device contrast is nominally significant but does not survive the family-wise correction over the three noise profiles, so the confirmatory evidence for the sign of the effect is the pooled analysis, and the per-profile values delineate its robustness across noise conditions.

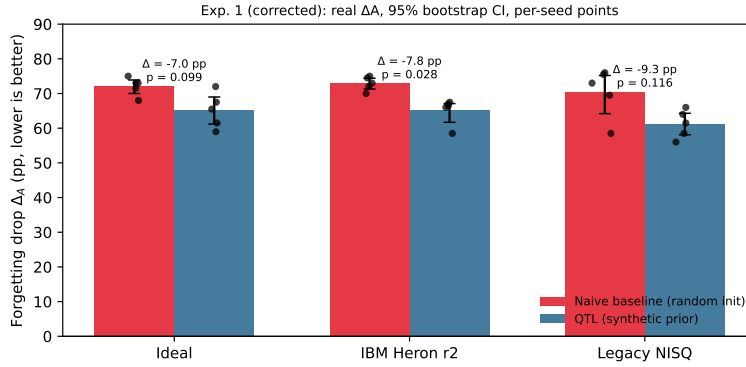


Figure 1: Experiment 1. Forgetting drop on Task A per noise profile for the random initialization and the synthetic prior. Bars show the mean over five seeds, markers show the per-seed values, and whiskers show the 95% bootstrap confidence interval of the paired reduction.

Table 3: Experiment 1 (corrected). Forgetting drop  $\Delta_A$  on Task A, in percentage points (mean  $\pm$  std over five seeds), with the paired reduction obtained by synthetic pre-training and its 95% bootstrap confidence interval. Significance from a two-sided paired  $t$ -test over seeds ( $df=4$ ).  $p_{\text{Holm}}$  applies the Holm procedure over the three per-profile tests treated as one family.  $d_z$  is the paired Cohen effect size. The Wilcoxon signed-rank test cannot reach  $p < 0.05$  with five seeds (minimum two-sided  $p$  is 0.0625). The QTL arm in this experiment runs Task A at  $\eta = 0.01$  and Task B at  $\eta = 0.005$ , against  $\eta = 0.05$  for both tasks in the baseline. the reduction therefore mixes the effect of the initialization with the effect of the optimization regime (see the learning-rate control experiment of the revision).

| Noise profile | Baseline $\Delta_A$ | QTL $\Delta_A$   | Reduction [95% CI] | $p$   | $p_{\text{Holm}}$ | $d_z$ |
|---------------|---------------------|------------------|--------------------|-------|-------------------|-------|
| Ideal         | $72.10 \pm 2.61$    | $65.10 \pm 5.09$ | 7.00 [1.10, 12.40] | 0.099 | 0.198             | +0.96 |
| Heron r2      | $72.90 \pm 2.01$    | $65.10 \pm 3.73$ | 7.80 [4.50, 12.30] | 0.028 | 0.084             | +1.51 |
| Legacy NISQ   | $70.50 \pm 7.19$    | $61.20 \pm 4.04$ | 9.30 [0.80, 17.10] | 0.116 | 0.198             | +0.90 |

The absolute magnitude of the effect deserves a precise statement. The reduction corresponds to roughly ten percent of the baseline drop under the device profile, and the residual loss of about sixty-five points shows that the synthetic prior delays the interference between tasks without cancelling it. The

learning rate used by the pre-trained arm in this experiment differs from the rate of the random reference, as the caption of Table 3 records, so the measured reduction mixes the effect of the initialization with the effect of the optimization regime. The extended protocol repeats the experiment with matched learning rates and adds regularization and replay baselines. That control is the subject of the next subsection, and the conclusions of this one are conditional on it.

### 6.2. Experiment 1 (controlled re-run): matched rates and continual-learning baselines

Table 4 reports the controlled re-run of Experiment 1 as a  $2 \times 2$  design that crosses the initialization of the quantum layer (scratch, synthetic prior) with the learning-rate regime of the sequential phases (the single rate  $\eta_A = \eta_B = 0.05$  of the conference protocol, or the gentler pair  $\eta_A = 0.01$ ,  $\eta_B = 0.005$  used by the pre-trained arm of Table 3), so that the initialization effect is observed at matched rates. The design is augmented with three continual-learning baselines applied to the same variational classifier: elastic weight consolidation with penalty strength  $\lambda \in \{10^2, 10^3, 10^4\}$  computed from the empirical Fisher diagonal at the Task A optimum, an experience-replay arm with a buffer holding 25% of the Task A training set, and a Dark Experience Replay++ arm with the same buffer. The campaign covers twenty seeds per cell under the ideal and Heron r2 noise profiles, 360 sequential trainings in total. Every stored run carries its absolute accuracies, the derived metrics and the identity of the executing node.

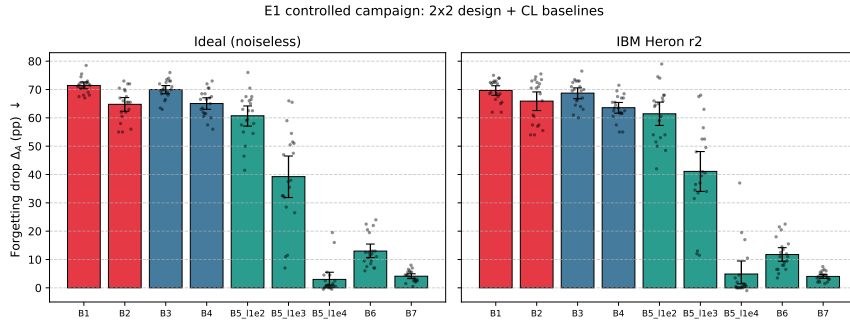


Figure 2: Experiment 1 (controlled re-run). Forgetting drop  $\Delta_A$  on Task A per arm under the ideal (left) and Heron r2 (right) noise profiles. The four cells of the  $2 \times 2$  design are shown in red (scratch initialization) and blue (synthetic prior). The continual-learning baselines are shown in green. Bars show the mean over twenty seeds with 95% bootstrap confidence intervals and markers show the per-seed values. Lower is better.

Table 5 decomposes the headline reduction of Table 3 into its two factors. The learning-rate regime carries most of the effect: at matched initialization, the gentler pair reduces the drop by 5.21 points from the scratch initialization (bootstrap interval  $[2.95, 7.51]$  for the reduction,  $p_{\text{Holm}} < 0.001$ ,  $d_z = -0.69$ ) and by 5.05 points from the synthetic prior ( $p_{\text{Holm}} < 0.001$ ,  $d_z = -0.85$ ) in

Table 4: Experiment 1 (controlled re-run): absolute accuracies and forgetting for the  $2 \times 2$  design (initialization  $\times$  learning-rate regime), strongly entangling ansatz, four qubits.  $\text{Acc}_A^{\text{init}}$ : Task A accuracy after the Task A phase.  $\text{Acc}_A^{\text{final}}$ : Task A accuracy after the Task B phase.  $\Delta_A = \text{Acc}_A^{\text{init}} - \text{Acc}_A^{\text{final}}$ .  $r_A = \text{Acc}_A^{\text{final}} / \text{Acc}_A^{\text{init}}$ . Mean  $\pm$  standard deviation over seeds. every run stores the executing node identity and the full per-seed payload.

| Arm                                                  | Pre-training / $\eta_A/\eta_B$ | $\text{Acc}_A^{\text{init}}$ | $\text{Acc}_A^{\text{final}}$ | $\text{Acc}_B$   | $\Delta_A$       | $r_A$             |
|------------------------------------------------------|--------------------------------|------------------------------|-------------------------------|------------------|------------------|-------------------|
| <i>Ideal (noiseless) (<math>n = 20</math> seeds)</i> |                                |                              |                               |                  |                  |                   |
| B1                                                   | none / 0.05/0.05               | $96.28 \pm 0.72$             | $24.85 \pm 2.78$              | $90.40 \pm 1.70$ | $71.42 \pm 2.79$ | $0.258 \pm 0.029$ |
| B2                                                   | none / 0.01/0.005              | $95.78 \pm 0.70$             | $31.00 \pm 5.67$              | $89.12 \pm 1.72$ | $64.78 \pm 5.72$ | $0.324 \pm 0.059$ |
| B3                                                   | synthetic / 0.05/0.05          | $96.25 \pm 0.68$             | $26.25 \pm 3.52$              | $89.67 \pm 1.78$ | $70.00 \pm 3.41$ | $0.273 \pm 0.036$ |
| B4                                                   | synthetic / 0.01/0.005         | $95.35 \pm 1.27$             | $30.30 \pm 4.67$              | $88.35 \pm 2.03$ | $65.05 \pm 4.78$ | $0.318 \pm 0.049$ |
| <i>IBM Heron r2 (<math>n = 20</math> seeds)</i>      |                                |                              |                               |                  |                  |                   |
| B1                                                   | none / 0.05/0.05               | $95.95 \pm 0.81$             | $26.27 \pm 3.68$              | $89.92 \pm 2.23$ | $69.67 \pm 3.97$ | $0.274 \pm 0.039$ |
| B2                                                   | none / 0.01/0.005              | $95.78 \pm 0.60$             | $29.88 \pm 7.76$              | $89.05 \pm 2.32$ | $65.90 \pm 7.79$ | $0.312 \pm 0.081$ |
| B3                                                   | synthetic / 0.05/0.05          | $96.00 \pm 0.92$             | $27.27 \pm 4.51$              | $89.88 \pm 1.74$ | $68.72 \pm 4.54$ | $0.284 \pm 0.047$ |
| B4                                                   | synthetic / 0.01/0.005         | $95.83 \pm 0.98$             | $32.25 \pm 4.21$              | $87.80 \pm 2.00$ | $63.58 \pm 4.45$ | $0.337 \pm 0.045$ |

the pooled analysis of the forty seed pairs. The initialization effect observed at matched rates is small and does not separate from zero: reductions of 1.19 points at the single rate ( $p_{\text{Holm}} = 0.309$ ) and 1.02 points at the gentle pair ( $p_{\text{Holm}} = 0.345$ ), with the same reading under each noise profile taken separately. The headline contrast between the pre-trained arm of the conference protocol and its scratch reference, 6.24 points with  $p_{\text{Holm}} < 0.001$  and  $d_z = -1.10$ , is therefore compatible with an effect of the optimization regime rather than of the prior, and the cross contrast reinforces this reading: the synthetic arm trained at the single rate forgets 4.03 points more than the scratch arm trained at the gentle pair ( $p_{\text{Holm}} = 0.004$ ), so the regime dominates the initialization even when the two factors are pitted against each other.

Table 6 places the synthetic prior against the continual-learning baselines, and on retention the contrast is not favourable to the prior. Relative to the plain scratch arm, the mildest consolidation reduces forgetting by 9.47 points and the replay-based methods by 58.19 (replay) and 66.49 points (Dark Experience Replay++), with  $p_{\text{Holm}} < 0.001$  for every contrast and  $|d_z|$  between 1.03 and 16.79, while the synthetic-prior arm forgets 3.24 points more than the mildest consolidation ( $p_{\text{Holm}} = 0.023$ ) and between 52 and 60 points more than the replay-based methods. These reductions are bought with plasticity rather than obtained for free. The strictest consolidation essentially blocks the second task: it ends the Task B phase with 91.08 to 93.30 points on Task A but only 31.55 to 34.12 points on Task B, against the 87.8 to 90.4 points reached by the plain and synthetic arms. Replay and Dark Experience Replay++ occupy intermediate positions on Task B (73.15 and 61.12 points under the ideal profile), so the table describes a stability-plasticity frontier rather than a ranking on  $\Delta_A$  alone.

The controlled evidence therefore narrows the claim that the conference version attached to Table 3: at this scale and with matched rates, the cross-domain

Table 5: Experiment 1 (controlled re-run): paired contrasts of the  $2 \times 2$  decomposition (initialization  $\times$  learning-rate regime) on the Task A forgetting drop  $\Delta_A$ , in percentage points (positive values mean more forgetting in the first arm of the contrast). Significance from a two-sided paired  $t$ -test over seeds.  $p_{\text{Holm}}$  applies the Holm procedure over the six pre-registered design contrasts treated as one family.  $d_z$  is the paired Cohen effect size. The pooled rows combine both noise profiles ( $n = 40$  seed pairs).

| Scope    | Contrast                     | $\delta$ [95% CI]    | $p$    | $p_{\text{Holm}}$ | $d_z$ | $n$ |
|----------|------------------------------|----------------------|--------|-------------------|-------|-----|
| Ideal    | LR effect (scratch): B2–B1   | −6.65 [−9.60, −3.88] | <0.001 | 0.001             | −0.99 | 20  |
|          | Init effect (high LR): B3–B1 | −1.43 [−3.42, +0.53] | 0.179  | 0.359             | −0.31 | 20  |
|          | LR effect (synth): B4–B3     | −4.95 [−7.50, −2.52] | 0.001  | 0.005             | −0.84 | 20  |
|          | Init effect (low LR): B4–B2  | +0.28 [−2.12, +2.62] | 0.827  | 0.827             | +0.05 | 20  |
|          | Headline (confounded): B4–B1 | −6.38 [−8.60, −4.08] | <0.001 | <0.001            | −1.19 | 20  |
|          | Cross contrast: B3–B2        | +5.22 [+2.15, +8.18] | 0.004  | 0.011             | +0.74 | 20  |
| Heron r2 | LR effect (scratch): B2–B1   | −3.77 [−7.33, −0.35] | 0.053  | 0.210             | −0.46 | 20  |
|          | Init effect (high LR): B3–B1 | −0.95 [−3.45, +1.45] | 0.474  | 0.474             | −0.16 | 20  |
|          | LR effect (synth): B4–B3     | −5.15 [−7.85, −2.60] | 0.001  | 0.007             | −0.83 | 20  |
|          | Init effect (low LR): B4–B2  | −2.33 [−5.75, +0.88] | 0.195  | 0.391             | −0.30 | 20  |
|          | Headline (confounded): B4–B1 | −6.10 [−8.68, −3.42] | <0.001 | 0.002             | −0.99 | 20  |
|          | Cross contrast: B3–B2        | +2.83 [−0.40, +6.17] | 0.115  | 0.346             | +0.37 | 20  |
| Pooled   | LR effect (scratch): B2–B1   | −5.21 [−7.51, −2.95] | <0.001 | <0.001            | −0.69 | 40  |
|          | Init effect (high LR): B3–B1 | −1.19 [−2.76, +0.38] | 0.154  | 0.309             | −0.23 | 40  |
|          | LR effect (synth): B4–B3     | −5.05 [−6.91, −3.26] | <0.001 | <0.001            | −0.85 | 40  |
|          | Init effect (low LR): B4–B2  | −1.02 [−3.15, +1.00] | 0.345  | 0.345             | −0.15 | 40  |
|          | Headline (confounded): B4–B1 | −6.24 [−7.95, −4.47] | <0.001 | <0.001            | −1.10 | 40  |
|          | Cross contrast: B3–B2        | +4.03 [+1.76, +6.26] | 0.001  | 0.004             | +0.55 | 40  |

prior contributes a small and non-significant reduction of forgetting under both noise conditions, the regime of the sequential phases contributes a significant one, and the regularization and rehearsal baselines absent from the conference version dominate retention at the explicit price of Task B plasticity.

### 6.3. Experiment 2: topology capacity and retention

The second experiment compares the three circuit topologies under the same sequential load. Table 7 reports, per topology, the Task A accuracy at the end of the Task A phase, the Task A accuracy at the end of the Task B phase, the resulting drop, the Task B accuracy and the relative retention. The corrected reading of this comparison differs from the one distributed with the conference version of this study, and the difference is instructive. The architectures with the largest capacity, SEL and TTN, reach the highest Task A accuracies, 96.5 and 95.4 points under the device profile, and both lose nearly all of that accuracy after the Task B phase, with retention values of 0.26 and 0.25. The basic entangler, whose single-axis rotations restrict its capacity, reaches only 73.1 points on Task A and retains 62.5 points, a retention of 0.86. The constrained topology appears the most stable in relative terms while being the weakest in absolute terms, since it never expresses the first task well enough to have much to lose. The paired tests resolve the contrast between the basic entangler and each of the other two topologies below the 0.001 level after Holm correction ( $p_{\text{Holm}} < 2 \times 10^{-4}$  under every profile), while the contrast between SEL and

Table 6: Experiment 1 (controlled re-run): continual-learning baselines against the plain scratch arm (B1) and the synthetic-prior arm (B4).  $\Delta_A$  is the Task A forgetting drop in percentage points (lower is better).  $p_{\text{Holm}}$  is the two-sided paired  $t$ -test over seeds after Holm correction within the family of ten baseline contrasts.  $d_z$  is the paired Cohen effect size of the contrast against B1 (negative values mean less forgetting than B1). EWC uses the empirical Fisher diagonal computed at the Task A optimum. Replay and DER++ replay a buffer with 25% of the Task A training set.

| Arm                                                  | Method              | $\Delta_A$        | $\text{Acc}_A^{\text{final}}$ | $\text{Acc}_B$    | vs. B1                         |          |
|------------------------------------------------------|---------------------|-------------------|-------------------------------|-------------------|--------------------------------|----------|
|                                                      |                     |                   |                               |                   | $\delta$ ( $p_{\text{Holm}}$ ) | $d_z$    |
| <i>Ideal (noiseless) (<math>n = 20</math> seeds)</i> |                     |                   |                               |                   |                                |          |
| B1                                                   | scratch             | $71.42 \pm 2.79$  | $24.85 \pm 2.78$              | $90.40 \pm 1.70$  | n/a                            | n/a      |
| B5-10 <sup>2</sup>                                   | EWC 10 <sup>2</sup> | $60.73 \pm 8.34$  | $35.55 \pm 8.14$              | $86.83 \pm 4.27$  | $-10.70$ ( $<0.001$ )          | $-1.21$  |
| B5-10 <sup>3</sup>                                   | EWC 10 <sup>3</sup> | $39.25 \pm 17.20$ | $57.02 \pm 16.97$             | $66.85 \pm 11.87$ | $-32.17$ ( $<0.001$ )          | $-1.82$  |
| B5-10 <sup>4</sup>                                   | EWC 10 <sup>4</sup> | $2.98 \pm 5.37$   | $93.30 \pm 5.15$              | $31.55 \pm 6.08$  | $-68.45$ ( $<0.001$ )          | $-12.08$ |
| B6                                                   | Replay              | $12.97 \pm 5.62$  | $83.30 \pm 5.52$              | $73.15 \pm 4.03$  | $-58.45$ ( $<0.001$ )          | $-9.60$  |
| B7                                                   | DER++               | $4.10 \pm 1.99$   | $92.17 \pm 1.89$              | $61.12 \pm 4.26$  | $-67.33$ ( $<0.001$ )          | $-17.44$ |
| B4                                                   | synth prior         | $65.05 \pm 4.78$  | $30.30 \pm 4.67$              | $88.35 \pm 2.03$  | n/a                            | n/a      |
| <i>IBM Heron r2 (<math>n = 20</math> seeds)</i>      |                     |                   |                               |                   |                                |          |
| B1                                                   | scratch             | $69.67 \pm 3.97$  | $26.27 \pm 3.68$              | $89.92 \pm 2.23$  | n/a                            | n/a      |
| B5-10 <sup>2</sup>                                   | EWC 10 <sup>2</sup> | $61.42 \pm 9.70$  | $34.52 \pm 9.38$              | $86.62 \pm 4.29$  | $-8.25$ (0.002)                | $-0.86$  |
| B5-10 <sup>3</sup>                                   | EWC 10 <sup>3</sup> | $41.10 \pm 16.72$ | $54.85 \pm 16.35$             | $70.55 \pm 9.85$  | $-28.57$ ( $<0.001$ )          | $-1.78$  |
| B5-10 <sup>4</sup>                                   | EWC 10 <sup>4</sup> | $4.88 \pm 9.50$   | $91.08 \pm 9.24$              | $34.12 \pm 10.09$ | $-64.80$ ( $<0.001$ )          | $-7.24$  |
| B6                                                   | Replay              | $11.75 \pm 5.57$  | $84.20 \pm 5.41$              | $73.03 \pm 3.66$  | $-57.92$ ( $<0.001$ )          | $-9.25$  |
| B7                                                   | DER++               | $4.03 \pm 1.67$   | $91.92 \pm 1.38$              | $60.85 \pm 3.45$  | $-65.65$ ( $<0.001$ )          | $-16.51$ |
| B4                                                   | synth prior         | $63.58 \pm 4.45$  | $32.25 \pm 4.21$              | $87.80 \pm 2.00$  | n/a                            | n/a      |

TTN does not reach the 0.05 level under any profile. The comparison across topologies is thus a comparison across effective capacities, which is consistent with the structural analysis of the readout noted in Section 4.

#### 6.4. Experiment 3: convergence under the cross-domain prior

The third experiment compares three initializations of the same model on the MNIST target, a random initialization, the synthetic prior and the MobileNetV2 prior, and Table 8 reports the resulting target accuracies. The three initializations reach accuracies inside a band of approximately 1.5 points, and no paired contrast reaches the 0.05 level under any profile. The synthetic prior does not raise the accuracy ceiling of the target task. The variance across seeds is smaller under the synthetic prior than under the random initialization in the device profile, 0.65 against 1.72 points, which indicates that the prior stabilises the outcome without changing its level. The convergence curves in Figure 5 show that the pre-trained arms start from lower cross-entropy values and reach the plateau earlier, which is consistent with the transfer-learning reading of the synthetic source.

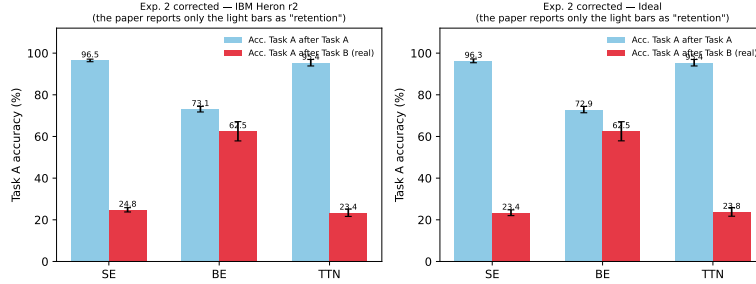


Figure 3: Experiment 2. Task A accuracy before and after the sequential exposure to Task B for each topology under the device noise profile. The gap between the two measurements is the true forgetting drop.

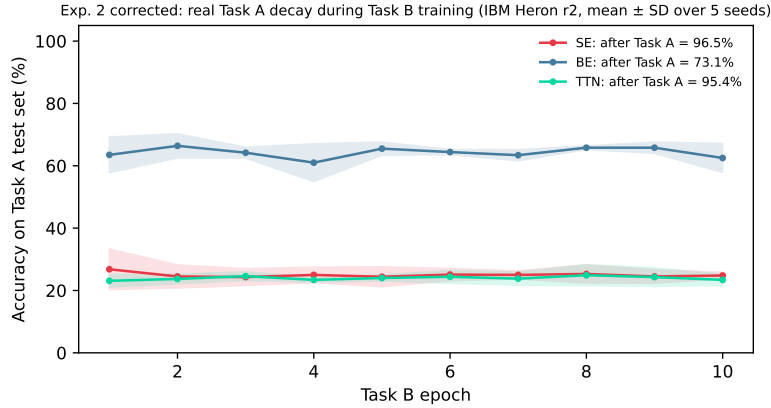


Figure 4: Experiment 2. Task A accuracy along the Task B phase for each topology, mean over five seeds with one standard deviation bands.

### 6.5. Scenario axis: task-IL and class-IL benchmarks

The scenario axis of the extended protocol asks whether the cross-domain prior behaves differently when the evaluation stops assuming a per-task readout. All runs of this axis train the shared multi-class head with globally offset labels and a single output layer, following the class-incremental protocol of van de Ven and Tolias [25]: at test time no task identity is available and the prediction is the argmax over the classes seen so far, while the task-incremental companion restricts the argmax to the task’s own pair of classes as an oracle. A single four-task chain (Fashion-MNIST{0,1} → MNIST{2,3} → KMNIST{4,5} → Fashion-MNIST{8,9}) therefore yields TIL-2, CIL-2, TIL-3, CIL-3 and CIL-4 from the same sequential training, with ten seeds per arm and the synthetic prior sharing the E1 pre-training checkpoint protocol.

Table 9 reports the chain. Under the ideal profile the four-task class-IL accuracy is 24.90 points for the scratch arm against 24.93 for the synthetic prior

Table 7: Experiment 2 (corrected). Task A accuracy before and after the sequential exposure to Task B, with the resulting forgetting drop, for the three ansätze. Mean  $\pm$  std over five seeds.  $\text{Acc}_A^{\text{init}}$  is measured at the end of the Task A phase and  $\text{Acc}_A^{\text{post}}$  at the end of the Task B phase, on the frozen Task A test set.

| Noise profile | Ansatz | $\text{Acc}_A^{\text{init}}$ | $\text{Acc}_A^{\text{post}}$ | $\Delta_A$ | $\text{Acc}_B$   | Retention |
|---------------|--------|------------------------------|------------------------------|------------|------------------|-----------|
| Ideal         | SE     | $96.30 \pm 0.91$             | $23.40 \pm 1.39$             | 72.90      | $90.20 \pm 1.60$ | 0.24      |
|               | BE     | $72.90 \pm 1.52$             | $62.50 \pm 4.60$             | 10.40      | $57.90 \pm 2.33$ | 0.86      |
|               | TTN    | $95.40 \pm 1.56$             | $23.80 \pm 2.08$             | 71.60      | $90.40 \pm 0.89$ | 0.25      |
| Heron r2      | SE     | $96.50 \pm 0.61$             | $24.80 \pm 1.04$             | 71.70      | $89.60 \pm 1.14$ | 0.26      |
|               | BE     | $73.10 \pm 1.34$             | $62.50 \pm 4.64$             | 10.60      | $58.20 \pm 2.36$ | 0.86      |
|               | TTN    | $95.40 \pm 1.56$             | $23.40 \pm 1.78$             | 72.00      | $89.80 \pm 0.67$ | 0.25      |
| Legacy NISQ   | SE     | $96.70 \pm 0.57$             | $24.60 \pm 3.23$             | 72.10      | $89.90 \pm 2.27$ | 0.25      |
|               | BE     | $72.30 \pm 0.91$             | $63.40 \pm 3.23$             | 8.90       | $58.30 \pm 1.25$ | 0.88      |
|               | TTN    | $95.30 \pm 1.20$             | $22.00 \pm 0.79$             | 73.30      | $90.60 \pm 0.74$ | 0.23      |

Table 8: Experiment 3. Target accuracy (percentage) on the MNIST 2/3 test set for the three initialization paradigms, with the paired difference of the synthetic prior against the scratch reference and its Holm-adjusted significance from a paired  $t$ -test over five seeds. No contrast falls below the 0.05 level in the current five-seed campaign.

| Noise profile | Scratch          | QTL-Syn          | QTL-Mob          | $\delta$ (Synn/aScr) | $p_{\text{Holm}}$ |
|---------------|------------------|------------------|------------------|----------------------|-------------------|
| Ideal         | $90.10 \pm 2.13$ | $90.90 \pm 1.60$ | $89.70 \pm 1.15$ | +0.80                | 1.000             |
| IBM Heron r2  | $90.80 \pm 1.72$ | $89.90 \pm 0.65$ | $88.60 \pm 1.52$ | -0.90                | 0.430             |
| Legacy NISQ   | $89.60 \pm 2.61$ | $89.90 \pm 1.39$ | $89.70 \pm 1.72$ | +0.30                | 1.000             |

(pooled over both noise profiles 24.91 vs 24.84, paired difference -0.07 points,  $p_{\text{Holm}} = 0.537$ ,  $d_z = -0.14$ ), and the shorter prefixes behave the same way: TIL-2 74.81 vs 74.40, CIL-2 44.94 vs 45.30, TIL-3 66.33 vs 63.65, CIL-3 27.83 vs 27.77, with every Holm-corrected contrast inside the pre-declared family of five scenario metrics staying above the 0.05 level. The class-incremental reading is markedly harder than the oracle reading: accuracies on the freshly trained task stay around ninety, while the earlier ones collapse to single digits, mostly to near zero (Figure 6), and, at this scale, the synthetic prior neither attenuates nor aggravates the loss in either regime. Under the Heron r2 profile the four-task accuracy is 24.93 against 24.75 (paired difference -0.17,  $p_{\text{Holm}} = 0.242$ ), so the null reading is stable across noise conditions.

The community-standard five-task benchmarks behave in the same direction. On split-MNIST (5 tasks  $\times$  2 classes, shared head) the average class-IL accuracy over the sequence is 18.97 points for the scratch arm and 19.02 for the synthetic prior ( $\delta = +0.05$ ,  $p_{\text{Holm}} = 0.240$ ,  $d_z = 0.40$ ), with average forgetting 77.42 against 77.44 points. The regularization and rehearsal arms land at 17.26 (EWC) and 34.45 (rehearsal) mean accuracy, with  $\delta = +15.49$  ( $< 0.001$ ) and -1.71 ( $< 0.001$ ) against scratch. A Friedman test over the four arms (10 seeds,



Figure 5: Experiment 3. Cross-entropy loss along the target-training epochs for the three initialization paradigms under the device noise profile. Lines show the mean over five seeds and the shaded bands show one standard deviation.

ideal profile) gives  $\chi^2 = 27.12$  ( $p = < 0.001$ ) with mean ranks scratch 2.40, synth 2.60, rehearsal 4.00, EWC 1.00 and a Nemenyi critical difference of 1.48 at  $\alpha = 0.05$  [37]. The rehearsal arm separates from scratch and EWC beyond the critical difference, while the synthetic prior remains statistically indistinguishable from scratch. Under the Heron r2 profile split-MNIST gives 19.06 against 19.01 points (0.480), and split-Fashion-MNIST 19.97 against 20.57 (0.204). Table 10 collects the four AA/AF/BWT/FWT metrics per arm and Figure 7 the retention of the first task along the five phases. Figure 8 reads the same benchmark at the level of the full accuracy matrices: the plain and prior arms collapse to zero on every earlier task, while rehearsal preserves about sixty percent of the first task after the whole sequence. The standard class-incremental benchmarks therefore do not reproduce the retention advantage that the two-task protocol of Experiment 1 attributes to the synthetic prior: the prior remains a neutral initialization at this scale, and the arms that explicitly trade plasticity for retention are the only ones that move the retention metrics, in the direction and at the plasticity cost already quantified in Table 6.

#### 6.6. Regularization family and interaction with the synthetic prior

The regularization family closes the comparison on the standard benchmarks. Table 11 reports elastic weight consolidation at the campaign strength, uniform L2 drift and synaptic intelligence at two penalty strengths each, and the two memory-based arms, all under the same protocol. Synaptic intelligence at the lower strength is the only weight-regularization baseline that improves average accuracy over the plain sequential model, with a paired difference of +4.07 points ( $p_{\text{Holm}} = 0.007$ ), and it does so while reducing average forgetting from 77.4 to 6.8 points, below every memory-based remedy in the comparison. At the higher strength the same algorithm freezes the model after the first task,

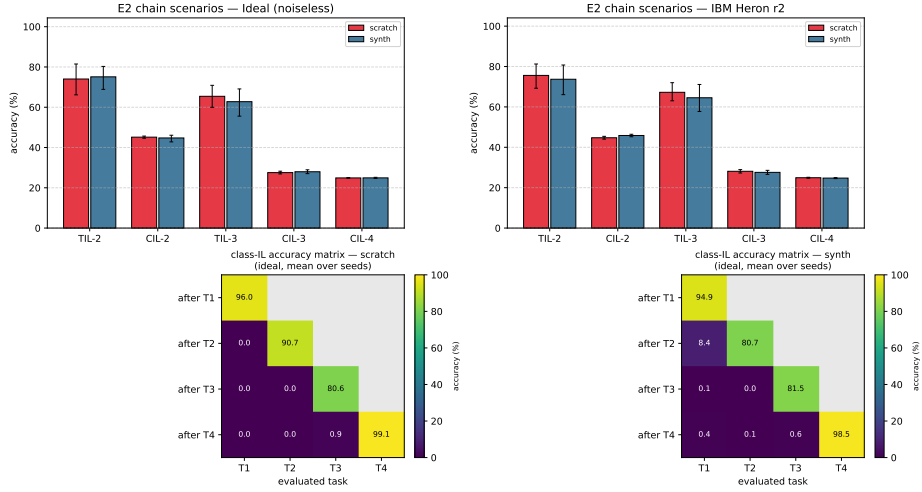


Figure 6: E2 scenario axis. Top: task-IL and class-IL accuracies at the two-, three- and four-task prefixes of the chain for the scratch and synthetic-prior arms ( $n = 10$  seeds, bars with 95% bootstrap intervals). Bottom: mean class-IL accuracy matrices after each phase (ideal profile). The upper triangle is not evaluated.

which shows the sensitivity of the family to the penalty calibration. Elastic weight consolidation and the uniform L2 penalty do not beat the plain baseline at either strength, with paired differences between  $-1.7$  and  $-6.7$  points. Dark experience replay reproduces the pattern of the two-task protocol on the five-task benchmark, with a paired gain of  $+11.78$  points ( $p_{\text{Holm}} < 0.001$ ) behind rehearsal ( $+15.49$  points) and ahead of every regularization arm.

The lower block of the table crosses every mechanism with the synthetic prior. Across the five method-matched contrasts, synthetic against random initialization under identical mechanism and penalty strength, the paired differences lie between  $-1.5$  and  $+1.9$  points with no  $p_{\text{Holm}}$  below  $0.18$ , and the plain contrast without a mechanism reproduces the null of the previous sections. Combining the prior with the strongest memory-free penalties therefore leaves the picture unchanged, and the synthetic initialization is inert for retention both alone and on top of the mechanisms that do act.

### 6.7. Longer task chains

The scenario and benchmark campaigns were complemented with ten-task chains, one class per task, on split-MNIST and split-Fashion-MNIST, to test whether the ranking of strategies survives longer sequences. Table 12 reports the average accuracy and average forgetting of the same arms under the same protocol. The plain model collapses to the ten-class chance level, with a forgetting of  $89.99$  points, and the synthetic prior does not change it, with a paired difference of  $+0.03$  points. The memory-based arms keep a significant advantage. Rehearsal gains  $+5.68$  points and dark experience replay  $+8.45$  points on

Table 9: E2 scenarios on the four-task chain (Fashion-MNIST{0,1}  $\rightarrow$  MNIST{2,3}  $\rightarrow$  KMNIST{4,5}  $\rightarrow$  Fashion-MNIST{8,9}), strongly entangling ansatz, four qubits. Task-IL (task oracle) and class-IL (argmax over classes seen so far) accuracies at the two-, three- and four-task prefixes of the same training run.  $\delta$  ( $p_{\text{Holm}}$ ): paired  $t$ -test of synth-scratch, Holm-corrected within the family of five scenario metrics per scope.  $d_z$  paired effect size. Mean  $\pm$  SD over seeds.

| Scenario                 | Scratch           | Synth             | $\delta$ [95% CI]     | $p_{\text{Holm}}$ | $d_z$ | $n$ |
|--------------------------|-------------------|-------------------|-----------------------|-------------------|-------|-----|
| <i>Ideal (noiseless)</i> |                   |                   |                       |                   |       |     |
| Task-IL 2                | 74.03 $\pm$ 13.02 | 75.10 $\pm$ 9.81  | +1.07 [−7.45, +9.15]  | 0.817             | +0.08 | 10  |
| Class-IL 2               | 45.15 $\pm$ 0.90  | 44.75 $\pm$ 2.92  | −0.40 [−2.38, +1.23]  | 0.692             | −0.13 | 10  |
| Task-IL 3                | 65.43 $\pm$ 9.20  | 62.77 $\pm$ 11.52 | −2.67 [−12.40, +6.67] | 0.616             | −0.16 | 10  |
| Class-IL 3               | 27.53 $\pm$ 1.15  | 27.97 $\pm$ 1.63  | +0.43 [−0.22, +1.15]  | 0.274             | +0.37 | 10  |
| Class-IL 4               | 24.90 $\pm$ 0.28  | 24.93 $\pm$ 0.48  | +0.03 [−0.35, +0.38]  | 0.901             | +0.04 | 10  |
| <i>IBM Heron r2</i>      |                   |                   |                       |                   |       |     |
| Task-IL 2                | 75.60 $\pm$ 10.33 | 73.70 $\pm$ 12.32 | −1.90 [−9.50, +5.42]  | 0.651             | −0.15 | 10  |
| Class-IL 2               | 44.73 $\pm$ 1.16  | 45.85 $\pm$ 1.04  | +1.12 [−0.05, +2.23]  | 0.098             | +0.58 | 10  |
| Task-IL 3                | 67.23 $\pm$ 7.84  | 64.53 $\pm$ 11.38 | −2.70 [−10.95, +5.92] | 0.569             | −0.19 | 10  |
| Class-IL 3               | 28.12 $\pm$ 1.45  | 27.58 $\pm$ 1.70  | −0.53 [−1.22, +0.17]  | 0.192             | −0.45 | 10  |
| Class-IL 4               | 24.93 $\pm$ 0.38  | 24.75 $\pm$ 0.40  | −0.17 [−0.45, +0.07]  | 0.242             | −0.40 | 10  |
| <i>Pooled</i>            |                   |                   |                       |                   |       |     |
| Task-IL 2                | 74.81 $\pm$ 11.47 | 74.40 $\pm$ 10.86 | −0.41 [−6.10, +5.19]  | 0.891             | −0.03 | 20  |
| Class-IL 2               | 44.94 $\pm$ 1.03  | 45.30 $\pm$ 2.21  | +0.36 [−0.84, +1.41]  | 0.544             | +0.14 | 20  |
| Task-IL 3                | 66.33 $\pm$ 8.37  | 63.65 $\pm$ 11.18 | −2.68 [−9.17, +3.78]  | 0.432             | −0.18 | 20  |
| Class-IL 3               | 27.83 $\pm$ 1.31  | 27.77 $\pm$ 1.63  | −0.05 [−0.58, +0.49]  | 0.861             | −0.04 | 20  |
| Class-IL 4               | 24.91 $\pm$ 0.32  | 24.84 $\pm$ 0.44  | −0.07 [−0.31, +0.15]  | 0.537             | −0.14 | 20  |

split-MNIST (Holm-corrected  $p < 0.001$  in both cases), and both gains reproduce on split-Fashion-MNIST at +7.28 and +8.77 points. Synaptic intelligence preserves its low-forgetting profile, 8.35 points against 89.99 of the plain model, but its accuracy advantage dilutes without significance, +1.08 points at the chosen strength and +0.66 and +0.76 points at the weaker ones, while elastic weight consolidation stays at the plain level at both tested strengths, −0.06 and −0.00 points. The hyperparameter probes of the same campaign confirm the insensitivity of the family. Doubling the rehearsal buffer from twenty-five to fifty percent changes nothing measurable, +0.22 points against the smaller buffer, and the elastic-weight penalty does not recover its deficit either, as it did not on the five-task benchmarks. Longer chains therefore sharpen the reading of the previous sections. The memory-free remedies do not carry retention when interference accumulates over ten tasks, and the arms with an explicit memory are the only components that improve it.

#### 6.8. Data scale: full binary splits and the sample-size curve

The data-scale axis re-runs the two-task binary protocol of Experiment 1 on the full class splits and on two reduced sample budgets, separating "small model" from "small data". Table 13 reports the per-seed means for 500, 2000

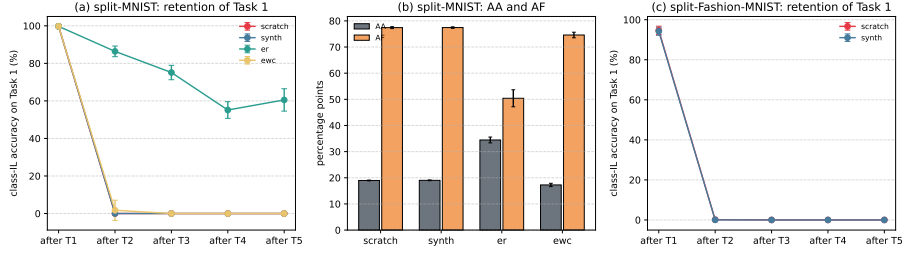


Figure 7: E2 class-incremental benchmarks. (a) Retention of the first task along the five phases of split-MNIST for the four arms, (b) AA and AF per arm, (c) retention of the first task for split-Fashion-MNIST. Means over ten seeds with one standard-deviation bands.

Table 10: E2 standard class-incremental benchmarks: split-MNIST and split-Fashion-MNIST (5 tasks  $\times$  2 classes, shared head, class-IL evaluation). AA/AF/BWT/FWT follow van de Ven et al. (2022).  $\text{Acc}_{t_0}$  is the accuracy on the first task after the whole sequence.  $\delta$  ( $p_{\text{Holm}}$ ): paired  $t$ -test vs. scratch on AA, Holm-corrected within each block.  $d_z$  paired effect size. Mean  $\pm$  SD over seeds.

| Bench.                             | Arm     | AA    | AF    | BWT    | FWT    | $\text{Acc}_{t_0}$ | $\delta$ AA [95% CI]    | $p_{\text{Holm}}$ | $d_z$  |
|------------------------------------|---------|-------|-------|--------|--------|--------------------|-------------------------|-------------------|--------|
| <i>split-MNIST (ideal)</i>         |         |       |       |        |        |                    |                         |                   |        |
|                                    | scratch | 18.97 | 77.42 | -96.77 | -13.78 | 0.00               | n/a                     | n/a               | n/a    |
|                                    | synth   | 19.02 | 77.44 | -96.80 | 0.00   | 0.00               | +0.05 [−0.02, +0.13]    | 0.240             | +0.40  |
|                                    | er      | 34.45 | 50.40 | -63.00 | -13.78 | 60.52              | +15.49 [+14.73, +16.10] | <0.001            | +13.20 |
|                                    | ewc     | 17.26 | 74.60 | -93.25 | -13.78 | 0.00               | −1.71 [−2.13, −1.28]    | <0.001            | −2.37  |
| <i>split-MNIST (Heron r2)</i>      |         |       |       |        |        |                    |                         |                   |        |
|                                    | scratch | 19.06 | 77.34 | -96.67 | -13.39 | 0.00               | n/a                     | n/a               | n/a    |
|                                    | synth   | 19.01 | 77.43 | -96.79 | 0.00   | 0.00               | −0.06 [−0.20, +0.05]    | 0.480             | −0.35  |
| <i>split-Fashion-MNIST (ideal)</i> |         |       |       |        |        |                    |                         |                   |        |
|                                    | scratch | 19.97 | 77.53 | -96.91 | -10.34 | 0.00               | n/a                     | n/a               | n/a    |
|                                    | synth   | 20.57 | 76.94 | -96.18 | 0.00   | 0.00               | +0.59 [−0.01, +1.51]    | 0.204             | +0.43  |

and 12 000 training samples per task using the shared binary readout of the two-task protocol: the mean accuracy over the two tasks is 85.70 / 85.72 points at 500 (scratch / synthetic prior), 62.19 / 62.79 at 2 000 and 63.75 / 62.04 at the full split, and the paired synth–scratch contrasts are +0.02 (0.968), +0.61 (0.356) and −1.71 (0.136) points (cf. Figure 9). Two readings survive the numbers: the shared-readout accuracy drops sharply from the 500-sample budget to the 2 000-sample budget and then plateaus, and the drop is carried entirely by Task-A retention: both tasks are still learned to the same level at every budget (freshly trained accuracies between 91 and 97 points) while the Task-A drop grows from 16.2 points at 500 samples to 64.9 and 60.2 at 2 000 and 12 000 (scratch arm), so the small budget of the conference protocol understated the interference between the two tasks rather than limiting the model, and the forgetting drop does not separate the arms at any budget, i.e. the null result of the controlled re-run is not an artefact of the small data subset. Under the Heron r2 profile the full

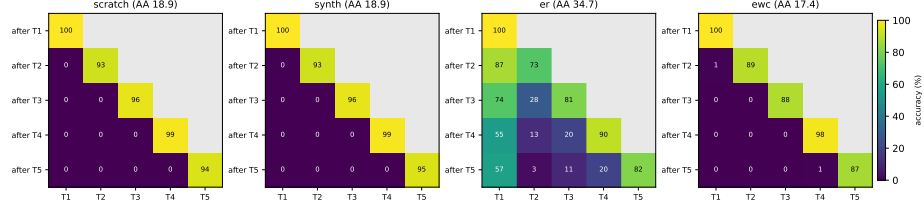


Figure 8: Class-incremental accuracy matrices of the four benchmark arms on split-MNIST, mean over ten seeds. The plain model and the synthetic prior lose every earlier task, while rehearsal preserves 60.5 percent of the first task and lifts the average accuracy by 15.5 points. Elastic weight consolidation does not retain at this model scale, which isolates the retention gain to the memory-based remedy.

split gives 62.32 / 62.11 points (0.825).

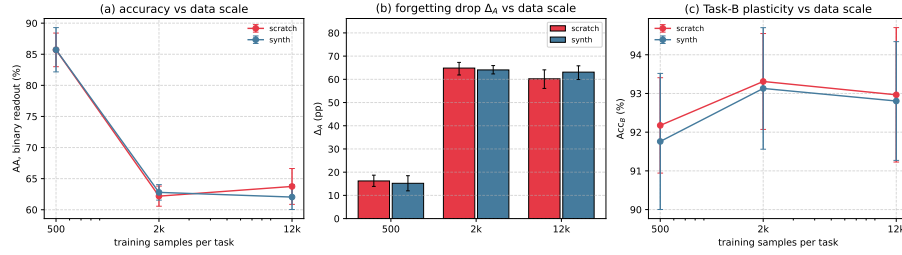


Figure 9: E3 data scale. (a) Mean accuracy over the two tasks of the shared binary readout, (b) Task-A forgetting drop, (c) Task-B accuracy, as a function of training samples per task (ideal profile, means over ten seeds with one standard-deviation error bars).

### 6.9. Scaling: qubits, depth and task count

The scaling axis sweeps the circuit resources of the two-task protocol (Table 14): qubit counts 4, 6, 8 with the PCA dimension matched to the number of qubits, and depths 2, 3, 4 at four qubits, ten seeds per cell. Across the grid the baseline four-qubit three-layer cell reaches a Task-A drop of 68.30 / 71.35 points (scratch / prior, ideal) and the paired synth–scratch contrasts remain inside noise in every configuration (4q/3L +3.05, 0.108, 6q/3L -0.85, 0.585, 8q/3L -0.55, 0.868, 4q/2L -2.45, 0.334, 4q/4L +0.00, 1.000). Adding qubits at fixed depth and changing depth at four qubits leave both the accuracy level and the forgetting profile within the seed spread (Figure 10), while the per-epoch elapsed time grows by roughly a factor of 2.3 from four to eight qubits at fixed depth, consistent with the state-vector simulation cost. The resource accounting of the same grid (parameter counts, one- and two-qubit gate counts, depth, and a fault-tolerant  $T$ -count estimate) is reported in Appendix Appendix A. The task-count dimension of the scaling question comes from the scenario chain

Table 11: E6 on the split-MNIST class-incremental benchmark (ideal profile, ten seeds). Top: weight-regularization baselines and memory-based arms, with the paired difference of average accuracy against the plain sequential baseline (95% bootstrap interval, Holm-corrected within the block). Bottom: interaction with the synthetic prior, with the paired difference against the same mechanism under random initialization. All values in accuracy points, mean  $\pm$  standard deviation over seeds.

| Arm                                                                                           | AA               | AF                | BWT                | $\Delta$ vs reference     | $p_{\text{Holm}}$ |
|-----------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|---------------------------|-------------------|
| <i>Regularization family and memory arms (<math>n = 10</math>)</i>                            |                  |                   |                    |                           |                   |
| scratch                                                                                       | $18.97 \pm 0.14$ | $77.42 \pm 0.37$  | $-96.77 \pm 0.47$  | $n/a$                     | $n/a$             |
| ewc                                                                                           | $17.26 \pm 0.62$ | $74.60 \pm 1.07$  | $-93.25 \pm 1.33$  | $-1.71 [-2.12, -1.27]$    | $< 0.001$         |
| l2 (0.2)                                                                                      | $12.23 \pm 1.94$ | $31.70 \pm 9.32$  | $-39.62 \pm 11.65$ | $-6.73 [-7.84, -5.63]$    | $< 0.001$         |
| l2 (2)                                                                                        | $14.29 \pm 4.36$ | $5.96 \pm 4.19$   | $-7.44 \pm 5.23$   | $-4.67 [-7.27, -2.19]$    | $0.008$           |
| si (5)                                                                                        | $23.03 \pm 3.30$ | $6.81 \pm 4.38$   | $-8.51 \pm 5.47$   | $+4.07 [+2.15, +5.97]$    | $0.007$           |
| si (50)                                                                                       | $19.95 \pm 0.04$ | $0.01 \pm 0.03$   | $-0.02 \pm 0.03$   | $+0.98 [+0.89, +1.09]$    | $< 0.001$         |
| derpp                                                                                         | $30.74 \pm 1.63$ | $27.41 \pm 2.93$  | $-34.27 \pm 3.66$  | $+11.78 [+10.75, +12.67]$ | $< 0.001$         |
| er                                                                                            | $34.45 \pm 1.12$ | $50.40 \pm 3.26$  | $-63.00 \pm 4.07$  | $+15.49 [+14.73, +16.10]$ | $< 0.001$         |
| <i>Interaction with the synthetic prior (<math>\Delta</math> vs same method, random init)</i> |                  |                   |                    |                           |                   |
| synth (plain)                                                                                 | $19.02 \pm 0.14$ | $77.44 \pm 0.29$  | $-96.80 \pm 0.36$  | $+0.05 [-0.02, +0.13]$    | $0.962$           |
| synth_si (5)                                                                                  | $22.16 \pm 4.40$ | $25.91 \pm 10.66$ | $-32.39 \pm 13.32$ | $-0.87 [-5.03, +2.95]$    | $1.000$           |
| synth_si (50)                                                                                 | $18.43 \pm 1.85$ | $18.82 \pm 1.69$  | $-23.53 \pm 2.11$  | $-1.51 [-2.49, -0.33]$    | $0.181$           |
| synth_l2 (0.2)                                                                                | $11.45 \pm 5.30$ | $8.57 \pm 5.27$   | $-10.71 \pm 6.59$  | $-0.79 [-4.58, +3.08]$    | $1.000$           |
| synth_l2 (2)                                                                                  | $16.14 \pm 3.86$ | $3.81 \pm 3.87$   | $-4.76 \pm 4.84$   | $+1.85 [-2.52, +5.91]$    | $1.000$           |
| synth_derpp                                                                                   | $29.39 \pm 1.22$ | $18.53 \pm 1.78$  | $-23.17 \pm 2.22$  | $-1.35 [-2.33, -0.33]$    | $0.181$           |

of Section 6.5, where the same sequential runs are read at two, three and four tasks. The combined reading is that within the simulated range the synthetic prior gives no measurable advantage in any corner of the grid, and the cross-domain effect reported for the conference protocol is specific to the two-task binary setting rather than a generic property of the method.

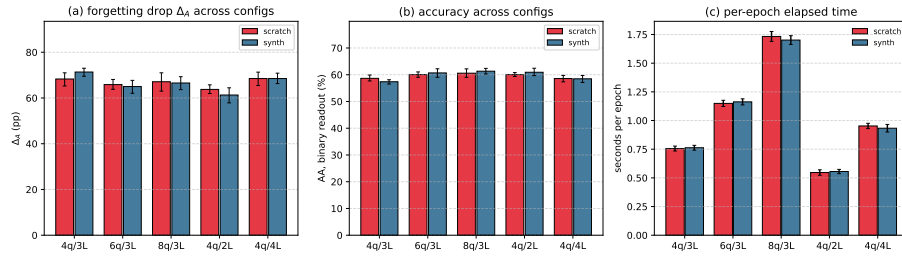


Figure 10: E4 scaling grid (ideal profile, ten seeds). (a) Task-A forgetting drop and (b) mean accuracy for the five configurations, (c) mean elapsed time per epoch on one CPU core.

#### 6.10. Capacity curve on the standard benchmarks

The scaling grid of the previous section uses the two-task protocol. Table 15 extends the sweep to the class-incremental benchmarks and to larger circuits,

Table 12: E7: ten-task chains, one class per task, class-incremental, ideal profile, ten seeds. Average accuracy and average forgetting of the main arms, with the paired difference of average accuracy against the plain model and its Holm-corrected probability within each chain. Mean  $\pm$  standard deviation over seeds.

| Arm                                                         | AA               | AF               | $\Delta$ vs scratch | $p_{\text{Holm}}$ |
|-------------------------------------------------------------|------------------|------------------|---------------------|-------------------|
| <i>split-MNIST, ten tasks (<math>n = 10</math>)</i>         |                  |                  |                     |                   |
| scratch                                                     | $10.01 \pm 0.01$ | $89.99 \pm 0.01$ | $n/a$               | $n/a$             |
| synth                                                       | $10.04 \pm 0.06$ | $89.96 \pm 0.06$ | +0.03               | 0.368             |
| si (5)                                                      | $11.09 \pm 1.21$ | $8.35 \pm 1.74$  | +1.08               | 0.116             |
| er                                                          | $15.69 \pm 0.72$ | $69.68 \pm 1.86$ | +5.68               | < 0.001           |
| derpp                                                       | $18.46 \pm 0.57$ | $49.39 \pm 1.41$ | +8.45               | < 0.001           |
| ewc ( $1 \times 10^4$ )                                     | $9.95 \pm 0.07$  | $89.76 \pm 0.17$ | -0.06               | 0.260             |
| <i>split-Fashion-MNIST, ten tasks (<math>n = 10</math>)</i> |                  |                  |                     |                   |
| scratch                                                     | $10.02 \pm 0.05$ | $89.97 \pm 0.05$ | $n/a$               | $n/a$             |
| synth                                                       | $10.03 \pm 0.05$ | $89.94 \pm 0.06$ | +0.01               | 0.665             |
| si (5)                                                      | $10.54 \pm 1.12$ | $9.03 \pm 1.27$  | +0.52               | 0.335             |
| er                                                          | $17.30 \pm 1.29$ | $70.51 \pm 3.06$ | +7.28               | < 0.001           |
| derpp                                                       | $18.79 \pm 0.77$ | $54.27 \pm 1.79$ | +8.77               | < 0.001           |

Table 13: E3 data scale: binary Fashion-MNIST $\{0, 1\} \rightarrow$  MNIST $\{2, 3\}$  protocol re-run on 500, 2000, 12000 training samples per task (full binary splits at 12k). The two-task protocol uses the shared binary readout of Experiment 1: AA is the mean accuracy over the two tasks of that readout and  $\Delta_A$  is the Task-A drop.  $\delta\text{AA}$  ( $p_{\text{Holm}}$ ): paired synth-scratch  $t$ -test, Holm within the three-size family.  $d_z$  paired effect size. Mean  $\pm$  SD over seeds.

| Samples     | Scratch AA       | Synth AA         | $\delta\text{AA}$ [95% CI] | $p_{\text{Holm}}$ | $d_z$ | Scratch $\Delta_A$ | Synth $\Delta_A$ |
|-------------|------------------|------------------|----------------------------|-------------------|-------|--------------------|------------------|
| 500         | $85.70 \pm 2.70$ | $85.72 \pm 3.56$ | +0.02 [-1.10, +0.96]       | 0.968             | +0.01 | 16.22              | 15.21            |
| 2k          | $62.19 \pm 1.62$ | $62.79 \pm 1.25$ | +0.61 [-0.64, +1.69]       | 0.356             | +0.31 | 64.85              | 64.05            |
| 12k         | $63.75 \pm 2.88$ | $62.04 \pm 2.00$ | -1.71 [-3.72, +0.13]       | 0.136             | -0.52 | 60.23              | 63.10            |
| 12k (Heron) | $62.32 \pm 2.84$ | $62.11 \pm 2.28$ | -0.21 [-1.93, +1.49]       | 0.825             | -0.07 | 62.72              | 63.19            |

with the number of qubits equal to the number of principal components of the encoding. Two effects carry decision weight. First, input capacity converts into accuracy only when a retention mechanism is present. From four to eight qubits the memory-based arms improve significantly, with paired gains of +5.9 points for rehearsal and +4.0 points for dark experience replay on the five-task chain and +4.8 and +2.9 points on the ten-task chain, while the plain model does not move and its forgetting stays at the level of the ten-task collapse. Second, the benefit saturates. Twelve qubits do not separate from eight in any arm, with paired differences of -1.0, +0.6 and -0.1 points for rehearsal, dark experience replay and the plain model, so the eight-qubit circuit is the largest size that pays off and the twelve-qubit cells mark a plateau rather than a further gain. Larger circuits leave classical state-vector simulation, sixteen qubits cost about forty-four seconds per optimisation step and thirty-two qubits require a state vector of sixty-nine gigabytes, and their discussion belongs with the fault-tolerant resource ladder of Appendix Appendix A.

Table 14: E4 scaling of the two-task protocol: qubits 4, 6, 8 and depth 2, 3, 4 (PCA dimension matched to the qubit count).  $\Delta_A$  (Task-A drop of the shared binary readout), AA and  $\text{Acc}_B$ .  $t_{\text{ep}}$  is the mean elapsed time per epoch (one CPU core, PennyLane state-vector).  $\delta\Delta_A$  ( $p_{\text{Holm}}$ ): paired synth-scratch  $t$ -test, Holm within the five-config family.  $d_z$  paired effect size. Mean over seeds.

| Config             | $\Delta_A$ scr. | $\Delta_A$ syn. | $\delta\Delta_A$ [95% CI] | $p_{\text{Holm}}$ | $d_z$ | AA scr. | AA syn. | $\text{Acc}_B$ scr. | $\text{Acc}_B$ syn. |
|--------------------|-----------------|-----------------|---------------------------|-------------------|-------|---------|---------|---------------------|---------------------|
| 4 qubits, 3 layers | 68.30           | 71.35           | +3.05 [+0.00, +6.25]      | 0.108             | +0.57 | 58.65   | 57.38   | 89.60               | 90.05               |
| 6 qubits, 3 layers | 65.85           | 65.00           | -0.85 [-3.80, +1.70]      | 0.585             | -0.18 | 60.05   | 60.67   | 90.95               | 90.55               |
| 8 qubits, 3 layers | 67.10           | 66.55           | -0.55 [-6.50, +5.25]      | 0.868             | -0.05 | 60.58   | 61.33   | 93.05               | 94.30               |
| 4 qubits, 2 layers | 63.75           | 61.30           | -2.45 [-7.25, +1.60]      | 0.334             | -0.32 | 60.05   | 60.92   | 89.85               | 89.45               |
| 4 qubits, 4 layers | 68.50           | 68.50           | +0.00 [-2.60, +2.90]      | 1.000             | +0.00 | 58.58   | 58.48   | 90.10               | 89.90               |

Table 15: E8/E9: capacity curve on the class-incremental benchmarks, ideal profile. Average accuracy (mean  $\pm$  standard deviation) against the number of qubits, equal to the number of principal components of the encoding. The memory-based arms gain significantly from four to eight qubits (rehearsal +5.9 and dark experience replay +4.0 points on the five-task chain, +4.8 and +2.9 on the ten-task chain, Holm-corrected  $p \leq 0.001$ ), while the plain model does not move, and twelve qubits do not separate from eight. The ten-task chain uses ten seeds, the five-task chain five (the twelve-qubit cell, five).

| Arm                                                 | 4 qubits         | 6 qubits         | 8 qubits         | 12 qubits        |
|-----------------------------------------------------|------------------|------------------|------------------|------------------|
| <i>split-MNIST, ten tasks (<math>n = 10</math>)</i> |                  |                  |                  |                  |
| scratch                                             | $10.01 \pm 0.01$ | $10.01 \pm 0.03$ | $10.00 \pm 0.02$ | <i>n/a</i>       |
| er                                                  | $15.69 \pm 0.72$ | $18.17 \pm 1.59$ | $20.46 \pm 1.64$ | <i>n/a</i>       |
| derpp                                               | $18.46 \pm 0.57$ | $20.69 \pm 0.88$ | $21.35 \pm 0.58$ | <i>n/a</i>       |
| <i>split-MNIST, five tasks (<math>n = 5</math>)</i> |                  |                  |                  |                  |
| scratch                                             | $19.01 \pm 0.12$ | <i>n/a</i>       | $19.31 \pm 0.28$ | $19.20 \pm 0.14$ |
| er                                                  | $34.49 \pm 1.27$ | <i>n/a</i>       | $40.35 \pm 2.25$ | $39.31 \pm 2.20$ |
| derpp                                               | $31.10 \pm 1.25$ | <i>n/a</i>       | $35.07 \pm 1.76$ | $35.66 \pm 1.42$ |

### 6.11. Mechanism analysis

The mechanism section combines the exploratory gradient probes of the review packet with the four checkpoint-based measurements of the extended protocol: per-layer gradient variance, empirical Fisher spectra, the loss barrier between the two task optima, and the parameter distributions including angle wrapping. All measurements use first-order autograd or double-precision finite differences with an explicit self-test, never second-order autodiff through the PennyLane layer, whose unreliability in this stack was documented during the audit (caveat T1). The checkpoints were re-trained locally for ten seeds per arm with the exact two-task protocol of the campaign (four-qubit strongly entangling ansatz, matched rates).

The gradient-variance probe of the review packet sweeps the barren-plateau proxy over qubit counts from four to twelve and both circuit topologies: the variance of the cost gradient decays from  $5.2 \times 10^{-3}$  to  $1.2 \times 10^{-5}$  for the strongly entangling ansatz between four and twelve qubits, while the hierarchical wiring decays far more gently, from  $6.8 \times 10^{-3}$  to  $3.6 \times 10^{-4}$ , keeping about thirty times

more gradient variance at twelve qubits [13, 28]. The checkpoint measurements sharpen this into a per-layer statement at the initialization the two arms actually use. At  $\theta_0$ , the pre-trained point, the per-layer gradient variance of the Task-A loss is 2.80e+00, 3.67e+00 and 2.09e+00 for the three ansatz layers (mean over layers 2.85e+00), against 2.73e-03, 3.03e-03 and 1.36e-03 at random initialization (2.37e-03), a gap of three orders of magnitude, uniformly across the layers, which means the source optimization lands the circuit in an active, high-gradient region of the landscape rather than the weak-gradient region that random rotation angles occupy (Figure 11a). After each arm has learned the first task the two checkpoints collapse onto comparably weak gradients (6.18e-02 vs 1.71e-01), and by the end of the second task both models exhibit renewed large gradients on Task-A data (1.15e+00 vs 2.12e+00), the signature of the interference that the accuracy matrices quantify.

The empirical Fisher spectrum at  $\theta_0$  tells the same story in terms of curvature: the trace is 1.13e+00 at random initialization against 1.27e+02 at the prior point, with a fuller spectrum (spectral entropy 1.05 vs 1.43 and effective rank 2.9 vs 4.2). The top eigenvalue carries about half of the trace in both cases (Figure 11b). The interpolation between the two task optima is the measurement that does not separate the arms: the linear path from  $\theta_A$  to  $\theta_B$  shows no loss barrier on the newly learned task for either arm (barrier height +0.000 and +0.000. The two task solutions are linearly connected in this parametrisation), and the parameter distance travelled is essentially identical ( $\|\theta_B - \theta_A\|/\sqrt{d} = 1.040$  vs. 1.063). Forgetting in this model is therefore not a barrier phenomenon between the two solutions. What separates the arms is where the optimization starts relative to both of them (Figure 11c).

Finally, the parameter distributions close an open item of the preliminary analysis: the circuit is  $2\pi$ -periodic in each rotation angle, which was verified numerically: wrapping the 36 angles of the pre-trained circuit reproduces the loss to floating-point precision (maximum  $|\Delta\mathcal{L}| = 3.3\text{e-}06$ ). Random initialization covers exactly one period (uniform in  $[0, 2\pi)$ ), while the source phase pushes some coordinates beyond it (raw range up to about 7.6 radians). After wrapping, the effective distributions differ mainly in dispersion (standard deviation 1.74 for random initialization against 2.01 for the prior, with 42% and 52% of the raw angles outside the canonical interval). The gradient and curvature evidence above is the part with decision weight. The distributional description documents the effective working point of the pre-trained circuit. Self-tests: the finite-difference check of the gradient path agrees at 6e-08 relative error, and the two ways of computing the Fisher trace agree at 2e-05.

## 7. Discussion

The results support a precise and bounded reading of synthetic cross-domain pre-training. The controlled factorial re-run redistributes the effect that the original protocol ascribed to the initialization. The apparent reduction of the forgetting drop, of about six points in the uncontrolled comparison, is carried by the learning-rate regime, which accounts for about five points under matched

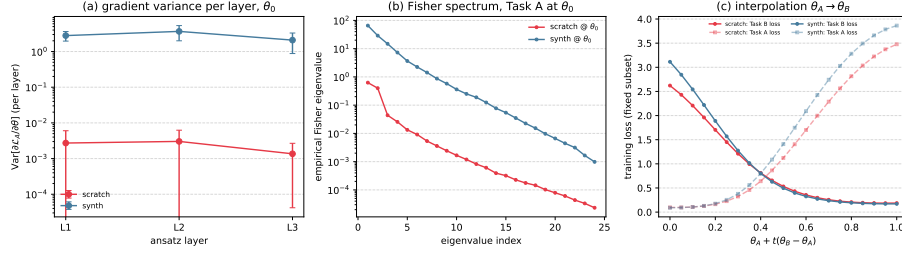


Figure 11: Mechanism measurements (10 seeds per arm). (a) Per-layer gradient variance of the Task-A loss at  $\theta_0$  (error bars: standard deviation over seeds), (b) mean empirical Fisher spectrum of the Task-A loss at  $\theta_0$ , (c) training losses along the linear interpolation between the two task optima (fixed 300-sample subsets).

schedules, while the synthetic initialization produces no significant change once the rates are matched. The benchmark campaign extends the null reading beyond the original scenario. The prior does not separate from a random initialization on the class-incremental sequences of up to five tasks, on the split benchmarks of MNIST and Fashion-MNIST, along the data budgets from five hundred to twelve thousand samples, or on the scaling grid over qubit counts and circuit depths. The study therefore answers its motivating question with a controlled null result at this scale, supported by an explicit statistical protocol and a full record of the per-seed payloads.

The mechanism evidence adds a second result. The pre-trained point is not a minimum of the target task and not a better starting region in terms of retention, yet its gradient signal is roughly three orders of magnitude stronger in the first layer than the signal available at a random initialization, and its empirical Fisher spectrum is about two orders of magnitude larger. The loss interpolation between the two task optima shows no barrier along the linear path, so the separation between the arms lies in the starting point rather than in the connectivity of the landscape. Taken together, the probes dissociate gradient signal from continual-learning performance at this scale, since a starting point with far more active gradients does not retain earlier tasks better. This dissociation is informative for the design of memory-free strategies because it removes gradient strength from the list of candidate mechanisms that alone can explain retention.

The component comparison completes the picture in the direction of practice. On the standard class-incremental benchmarks, rehearsal yields the largest benefit, with an advantage of about fifteen points over a plain sequential baseline, at a quantified cost in plasticity, while elastic weight consolidation trades most of its plasticity for the retention gains it obtains. Among the weight-regularization baselines, synaptic intelligence is the only one that improves over the plain model, with a modest gain of about four points and a tenfold reduction of forgetting, which places the regularization family behind the memory-based remedies in this comparison. The synthetic prior, by contrast, behaves as a neu-

tral initialization under this protocol. For practitioners, the operational reading is that memory-free pre-training can be used without harm but should not be expected to carry retention, and that claims of transfer effects in quantum continual learning should be established under a matched optimization regime before attribution.

Three constraints delimit the scope of the claims. The first is scale. The headline experiments use four qubits while the scaling grid reaches eight, the sequences contain up to five tasks, and the noise model, while calibrated on a current device generation, is simulated rather than executed on hardware. The second is the evaluation scenario. The extended protocol now covers the class-incremental setting of the classical benchmarks [25] together with the task-incremental setting of the original protocol, and the null reading persists in both. The third is the classical share of the pipeline. The MobileNetV2 feature extraction and the principal-component reduction dominate the processing cost of the image sources, which bounds any practical advantage that the quantum head could claim in the current configuration.

Several directions follow from these limits. A validation on physical hardware would test whether the dissociation between gradient strength and retention survives realistic noise correlations beyond the calibrated model. Longer task sequences and stronger feature spaces would probe whether the null reading persists as interference accumulates further, which the present budgets begin to sketch. Meta-learned initializations, which optimize the starting point for sequential adaptation rather than for a single source task, are a natural candidate for a strategy that acts where the mechanism probes look, and their evaluation under this protocol is left for future work.

## 8. Conclusions

This study characterized cross-domain synthetic pre-training as a memory-free initialization for sequential quantum classifiers through a controlled factorial design and a benchmark campaign of 650 training runs. The controlled evidence relocates the effect that the original protocol ascribed to the initialization. Under matched learning rates the synthetic prior produces no significant change in forgetting, while the optimization regime carries a reduction of about five points, and the null reading of the initialization is stable across class-incremental benchmarks of up to five tasks, qubit counts from four to eight and data budgets from five hundred to twelve thousand samples. The corrected reading of the topology comparison further shows that the apparent retention advantage of the constrained architecture reflects its limited capacity to learn the first task. The mechanism probes show that the prior places the circuit in a much more active gradient region without a corresponding gain in retention, so gradient signal and continual-learning performance dissociate at this scale. Among the evaluated remedies, rehearsal is the component with the largest measured benefit on the standard benchmarks, at a quantified cost in plasticity, and synaptic intelligence is the only weight-regularization baseline that improves over the plain

model while reducing forgetting by an order of magnitude. The evidence is limited to simulated devices and circuits of at most eight qubits, and the classical front-end cost of the pipeline is declared explicitly. The practical reading is twofold: claims of transfer effects in quantum continual learning must be established under matched optimization regimes, and memory-free initialization alone does not remove forgetting at this scale.

## Appendix A. Circuit resources and fault-tolerant estimate

Table A.16 lists the gate-level resources of the trained ansatz for the qubit counts and depths of the scaling grid. Every rotation gate of the circuit is a general single-qubit rotation. Under the standard synthesis cost model of Ross and Selinger [38] an arbitrary  $z$ -rotation approximated to error  $\varepsilon$  costs about  $3 \log_2(1/\varepsilon)$   $T$  gates, and no generic single-qubit gate can be synthesised below three  $T$  gates [39]. At  $\varepsilon = 10^{-6}$  the estimate is about 60  $T$  gates per rotation, i.e. about 2392  $T$  for the four-qubit three-layer circuit and 4784  $T$  at eight qubits. CNOTs are Clifford operations and contribute no  $T$  gates. The estimate counts every Euler angle of each rotation gate separately and is therefore conservative at the gate level. It excludes state preparation, measurement, and error-correction overheads, and it is reported as an order-of-magnitude reality check on what the demonstrated scale implies for the fault-tolerant era, not as a compilation study. At twelve qubits and four layers the same accounting reaches 9328  $T$  gates per inference, which is the quantitative form of the resource-awareness argument in the introduction.

## CRedit authorship contribution statement

**D. Martín-Pérez:** conceptualisation, methodology, software, investigation, writing. **F. Rodríguez-Díaz:** methodology, validation, writing, review and editing. **D. Gutiérrez-Avilés:** validation, formal analysis, review and editing. **A. Troncoso:** supervision, funding acquisition, review and editing. **F. Martínez-Álvarez:** supervision, funding acquisition, conceptualisation, review and editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The simulation pipeline, the per-seed result payloads and the statistical analysis scripts are available in the public repository at [https://github.com/danimap27/Experimento\\_CrossDomain\\_QTL](https://github.com/danimap27/Experimento_CrossDomain_QTL), which also contains the job lists, the analysis code and the per-seed payloads of the regularization-family study of Section 6.6.

Table A.16: E4 appendix: circuit resources of the trained ansatz (AngleEmbedding  $R_x$  + strongly entangling layers +  $Z$  readout on wires 0–1), noiseless, and a fault-tolerant cost estimate. *Params* is the number of trainable rotation parameters of the VQC. the classical head adds  $2k+k$  parameters for  $k$  classes.  $T$  is the estimate assuming Ross–Selinger synthesis of each of the  $3n_q n_\ell + n_q$  Euler rotations at synthesis error  $\varepsilon = 10^{-6}$  ( $T_{\text{rot}} \approx 3 \log_2(1/\varepsilon)$ ). the floor uses the generic single-qubit-gate lower bound of 3  $T$  gates per rotation (Bocharov et al.). CNOTs are Clifford and contribute no  $T$  gates.

| $n_q$ | $n_\ell$ | Params | 1q (Rot+Rx) | 2q (CNOT) | Depth | $T$ floor | $T$ est. ( $\varepsilon=10^{-6}$ ) |
|-------|----------|--------|-------------|-----------|-------|-----------|------------------------------------|
| 4     | 2        | 24     | 12          | 8         | 9     | 84        | 1674                               |
| 4     | 3        | 36     | 16          | 12        | 14    | 120       | 2392                               |
| 4     | 4        | 48     | 20          | 16        | 18    | 156       | 3109                               |
| 6     | 2        | 36     | 18          | 12        | 12    | 126       | 2511                               |
| 6     | 3        | 54     | 24          | 18        | 15    | 180       | 3588                               |
| 6     | 4        | 72     | 30          | 24        | 19    | 234       | 4664                               |
| 8     | 2        | 48     | 24          | 16        | 15    | 168       | 3349                               |
| 8     | 3        | 72     | 32          | 24        | 19    | 240       | 4784                               |
| 8     | 4        | 96     | 40          | 32        | 22    | 312       | 6219                               |
| 10    | 2        | 60     | 30          | 20        | 18    | 210       | 4186                               |
| 10    | 3        | 90     | 40          | 30        | 23    | 300       | 5979                               |
| 10    | 4        | 120    | 50          | 40        | 26    | 390       | 7773                               |
| 12    | 2        | 72     | 36          | 24        | 21    | 252       | 5023                               |
| 12    | 3        | 108    | 48          | 36        | 26    | 360       | 7175                               |
| 12    | 4        | 144    | 60          | 48        | 30    | 468       | 9328                               |

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